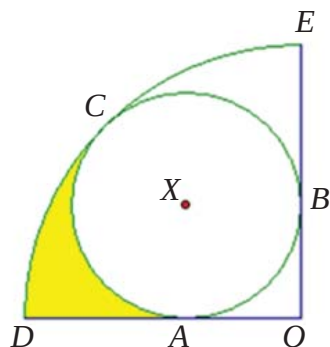


- 3 *ODE* is a quadrant with centre *O* and radius 6 cm.
ABC is a circle with centre *X* which touches the sides of the quadrant *ODE* at the points *A*, *B* and *C*.



- (a) Show that the radius of the circle *ABC* is 2.49 cm, correct to 3 significant figures. [3]

- (b) Find the length of the minor arc *AC*.

Answer _____ cm [3]

- (c) Find the area of the shaded region.

Answer _____ cm² [3]

- 4 (a) It is given that $\mathcal{E} = \{x : x \text{ is a real number such that } 10 \leq x \leq 30\}$,
 $A = \{x : x \text{ is a multiple of } 4\}$,
 $B = \{x : x \text{ is a multiple of } 3\}$ and
 $C = \{x : x \text{ is a multiple of } 12\}$.

- (i) List all the elements in B .

Answer $B =$ _____ [1]

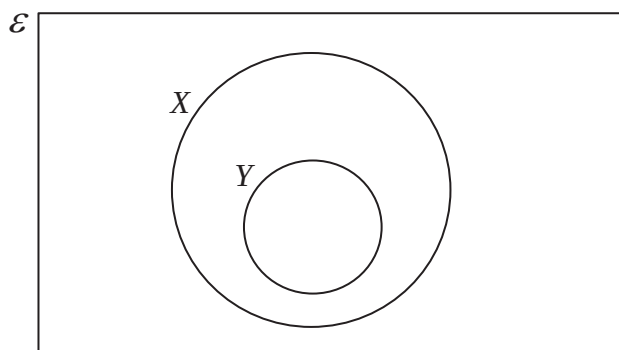
- (ii) Find $n(A' \cap B')$.

Answer _____ [1]

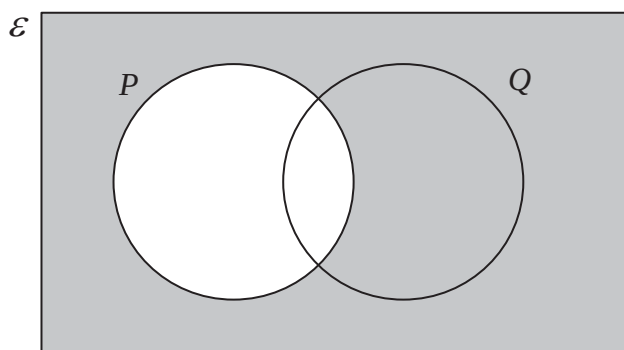
- (iii) Write down, in set notation, the relationship between sets A , B and C .

Answer _____ [1]

- (b) (i) On the Venn diagram shown in the answer space, shade the set $X' \cup Y$. [1]

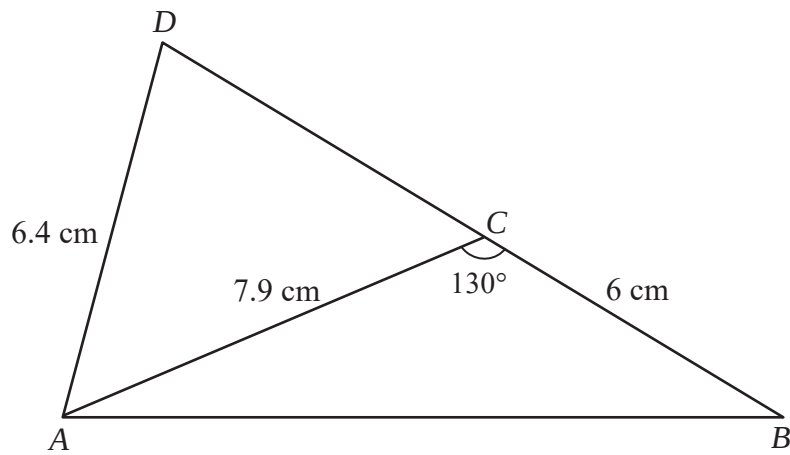


- (ii) Use set notation to describe the set shaded in the Venn diagram below.



Answer _____ [1]

5



BCD is a straight line.

$AC = 7.9$ cm, $AD = 6.4$ cm and $BC = 6$ cm.

Angle $ACB = 130^\circ$.

(a) Calculate AB .

Answer _____ cm [3]

(b) Calculate angle ADC .

Answer _____ [2]

(c) Calculate the shortest distance from C to AB .

Answer _____ cm [3]